

```

scher@pe15:~$ cat 3684.txt
First name: Sophia
Last Initial: C

Preferred skills: Python, java, frontend, web frameworks, (React/Flask), ML, data science
Preferred roles: presentation developer
scher@pe15:~$ █

```

1.
2. .

- a. The essential characteristic of a distributed version control system is the presence of a mirror copy of the shared repository on each group member's computer. This allows for the repository to be recovered if the copy stored on the server computer is lost.
- b. The advantage of the snapshot-based approach is that it allows for easier recovery of lost older versions, since the entire file layout is saved, unlike other non-Git VCS.

```

scher@pe15:~$ git config --global user.name "Sophia C"
scher@pe15:~$ git config --global user.email sophia.cherkaoui001@umb.edu
scher@pe15:~$ git config --list
user.name=Sophia C
user.email=sophia.cherkaoui001@umb.edu
scher@pe15:~$ █

```

- c.
- d. When a file is stored, Git generates a checksum in the form of a SHA-1 hash. Then, the file is referenced using its checksum value, which means that any changes to the file will result in a miscalculation of the old checksum, and thus Git is made aware of the change.
- e. The three main Git workflow steps are: modified, staged, and committed.
- f. Checksum: `55659dcca3279c34e31382f6d7ad1f635d70043`

3. .

- a. The complex developer environment is one where the “probe, sense, respond” framework is used. It emphasizes the analysis and full understanding of a problem or goal before developing a solution. It promotes a “fail forward” mindset where mistakes are learned from and are used to identify potential systemic issues. Because of the educational aspect of this environment, complex developer teams highly prioritize communication.
- b. .
 - i. 2-1: The Scrum practices can be divided into four main areas: roles (ex, product owner), activities (ex, sprints), artifacts (ex, product backlog), and rules.
 - ii. 2-2: A scrum team consists of a product owner (1 person), scrum master (1 person), and development team (usually several people).
 - iii. 2-3: The Scrum framework has several milestones, beginning with dividing tasks into a product backlog, continuing with sprint-related steps, and ending with an incrementally shippable product and reflections.
 - iv. 2-4: The product backlog provides a clear organization of project tasks in order of high to low priority, allowing for the completion of some tasks before others.

- v. 2-5: A Scrum team works to develop the product backlog by prioritizing certain tasks, estimating time/resources required, and finally refining their list of tasks.
- c. My team's website project included the following features: a large title header with button links, as well as the contents of the main page. These features were split into parts as follows:
 - i. Title header: (1) code HTML layout/header width, (2) add button links to members' websites, (3) add CSS styling of header/button text
 - ii. Contents of main page: (1) HTML layout of text (i.e., how much of the webpage space it will use), (2) adding larger text for members' names, (3) adding email links for each member, and (4) adding CSS styling.
 - iii. Problems encountered: One difficulty was working on the header layout for the webpage, since it requires knowledge of HTML web development techniques that not everyone on the team knew beforehand. Additionally, since this was our team's first project, we had to work on getting into the scrum mindset when working together. This project taught us about the challenge of working with individuals of varying levels of web development skills, as it required splitting the product backlog into tasks that the entire team could work towards with a minimal learning curve.